LCA CASE STUDY · AALTO UNIVERSITY · ECO-DESIGN & LIFE CYCLE ASSESSMENT

Eco-Design & Carbon Neutrality Roadmap

A full life-cycle assessment of an off-grid residential solar power system — panels, inverter, and battery storage — built into a practical roadmap toward carbon neutrality.

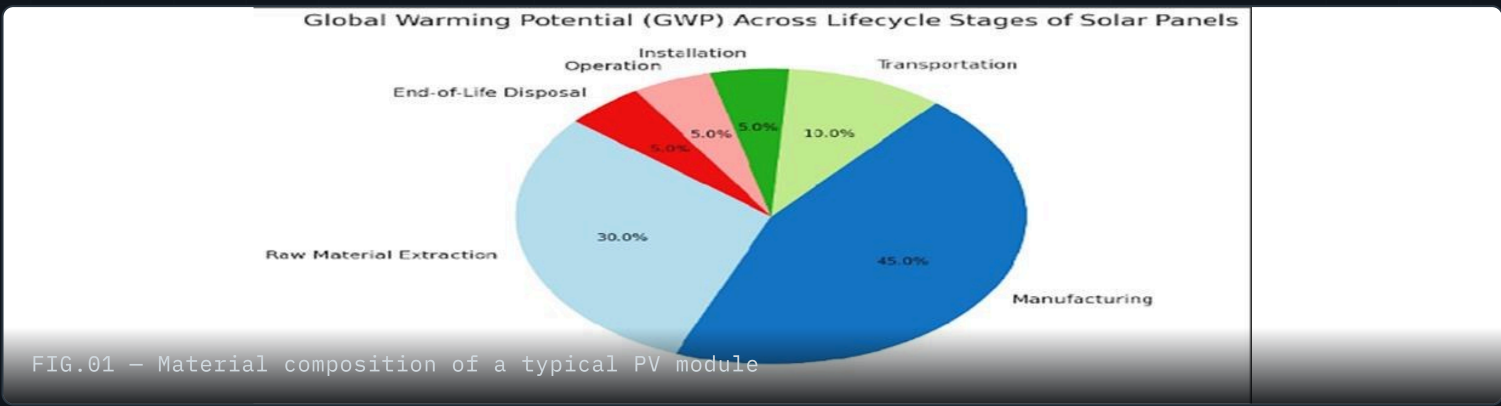

FIG.01 — Material composition of a typical PV module

3

SUBSYSTEMS MODELED END TO END

67.5 → 22.7

GWP RANGE ACROSS PANEL TECH (GCO_2/KWH)

94%

RECYCLED ALUMINIUM IN THE INVERTER

1.2 yrs

INVERTER CARBON PAYBACK TIME

01 WHY ECO-DESIGN THE GRID EDGE

Solar panels are marketed as a fuel-free, low-carbon energy source — and over their lifetime they are. But a **Life Cycle Assessment (LCA)** looks past the use phase to the full picture: raw material extraction, manufacturing, transport, operation, and end-of-life. For a residential off-grid system, that means the panels are only one part of the story — the inverter and battery storage carry their own, often underappreciated, carbon footprint.

Our team modeled a residential off-grid PV system as three linked subsystems — **PV panels, inverter, and battery storage** — and combined the findings into a roadmap toward carbon neutrality.

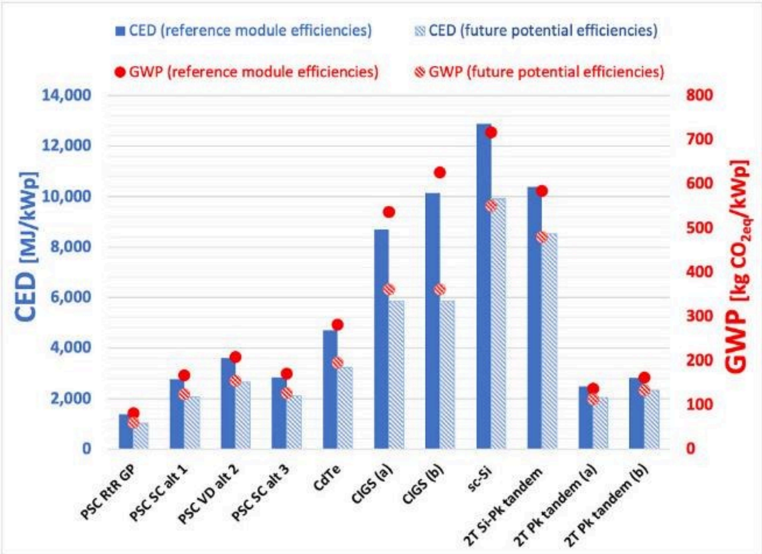

FIG.02 — Cradle-to-grave life cycle phases used across all three subsystem assessments

My focus was the **PV panel technology review** — comparing published LCA studies across silicon and thin-film cell types — plus building the **roadmap mindmap** that ties all three subsystems together, and coordinating the team's presentation.

02 PV PANEL TECHNOLOGIES

Thin-film beats silicon on carbon, not on efficiency

Pulling together dozens of published LCA studies, a clear pattern emerges: crystalline silicon panels (mono- and multi-Si) still dominate the market and offer the highest efficiencies, but thin-film technologies — CdTe and the chalcogenides (CIS/CIGS/CZTS) — have a meaningfully smaller carbon footprint and pay back their embodied energy faster.

MONO-SI

Monocrystalline silicon

Highest efficiency, most energy-intensive to produce

EPBT 3.0 yr

67.5 gCO_2/kWh

MULTI-SI

Multicrystalline silicon

Lower efficiency than mono-Si, somewhat lower GWP

EPBT 2.8 yr

48.4 gCO_2/kWh

CIS / CIGS / CZTS

Chalcogenide thin-film

Thin-film, lower material use, competitive efficiency

EPBT 1.41 yr

23.9 gCO_2/kWh

CDTE

Cadmium telluride thin-film

Lowest GWP and EPBT of commercial technologies

EPBT 1.32 yr

22.74 gCO_2/kWh

Global warming potential, gCO_2/kWh (scale relative to mono-Si)

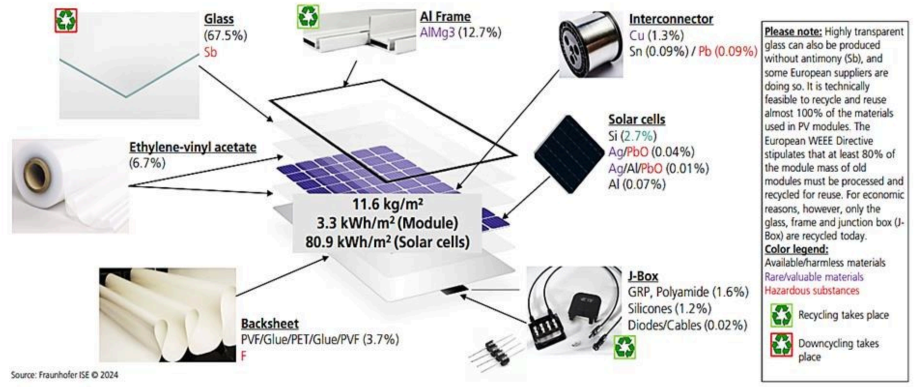

FIG.03 — Manufacturing (45%) and raw material extraction (30%) drive three-quarters of panel GWP

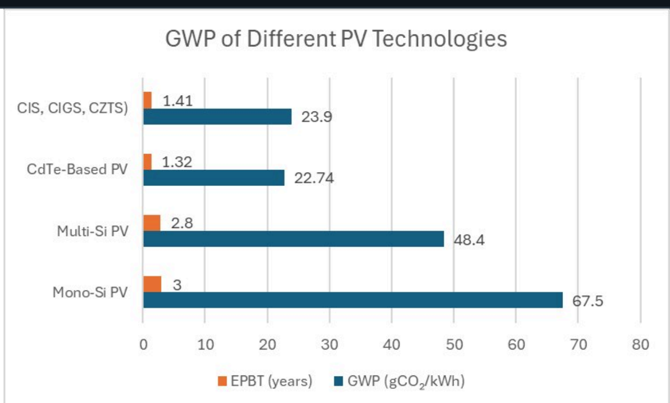

FIG.04 — GWP and energy payback time (EPBT) across commercial PV technologies

Why published LCAs vary so much

Across the literature we reviewed, GWP ranged from **10–194 gCO_2/kWh** for the same broad technology. The spread traces back to four factors: how much of the system boundary a study includes (mining and end-of-life vs. production only), cell efficiency, regional solar irradiance, and how recent the underlying manufacturing data is.

End-of-life is the weak link

Recycling recovers most of a panel's glass (70% of mass) and aluminium frame, but the industry still lacks consistent data and infrastructure for end-of-life management. Cumulative PV waste is projected to reach **~7 million tonnes by 2050** under an early-loss scenario — designing panels for disassembly (e.g. replacing EVA encapsulant with a removable thermoplastic) is one of the clearest levers left.

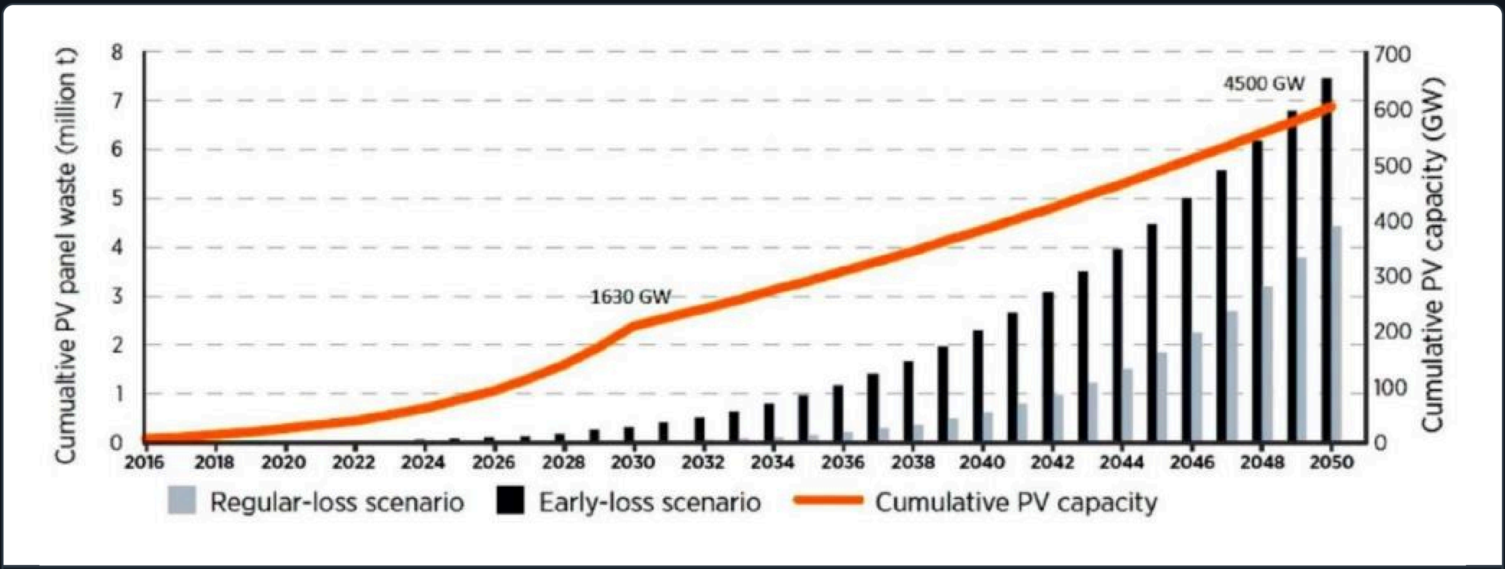

FIG.05 — Projected cumulative PV waste vs. installed capacity through 2050

03 INVERTER

A 954 kg CO_2e footprint that pays for itself in 14 months

Power electronics rarely get the eco-design attention panels do, so the team ran a full cradle-to-grave LCA on a real product: the Fronius Symo GEN24 Plus 10.0, a 2,547-part, 423-component residential inverter, using a bill-of-materials built up component by component.

954 kg

Total CO_2e over a 20-year lifetime

12.7%

Share of total PV system impact

1.2 yrs

CO_2 payback time

15,986 kg

CO_2e avoided over its lifetime

FIG.06 — GEN24 Symo 10.0 inverter, disassembled for bill-of-materials analysis

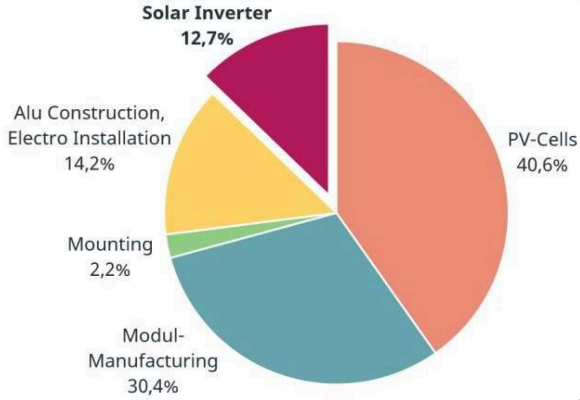

FIG.07 — Si-IGBT vs. SiC-MOSFET power modules: smaller die, lower module footprint, pricier wafer

Night consumption dominates

47.2% of impact

The inverter draws standby power even when the sun isn't shining — for data management and monitoring. On Germany's carbon-intensive grid (519 g CO_2e/kWh, ~30% coal), that standby draw alone accounts for **450 kg CO_2e**, just under half the inverter's lifetime footprint. Switching that draw to green electricity cuts the inverter's total footprint from 954 kg to just 38 kg CO_2e.

SiC helps the module, hurts the wafer

A SiC-MOSFET power module has less than half the footprint of an equivalent Si-IGBT module (11.04 vs. 26.44 kg CO_2e) thanks to smaller die sizes and lighter packaging. But SiC wafer production itself is far more energy-intensive per m² (90,000 vs. 27,300 kg CO_2e) — the module-level win comes from using less of a more carbon-intensive material.

Recycled aluminium, real savings

80 → 48.1 kg

Aluminium makes up nearly a third of the inverter's weight and 16% of its footprint. Using 94% recycled aluminium for the housing and heatsink — rather than primary stock — cuts that component's emissions nearly in half.

Where the data runs thin

Manufacturers rarely publish full environmental data for their products, so component-level LCI data had to come from statistical databases — with real spread between sources (e.g. ceramic capacitor footprints ranged 82.7–279.4 kg CO_2e/kg depending on the database used). More primary data from manufacturers would materially improve LCA accuracy here.

04 BATTERY STORAGE

Cell chemistry matters less than the box it comes in

A domestic storage system (DSS) buffers a PV system's daily mismatch between generation and demand — charging by day, discharging at night. The team compared four cell chemistries (LFP, sodium-ion, and two nickel-manganese-cobalt blends) across manufacturing, use, and end-of-life impacts, per kWh delivered over a 20-year, ~7,300-cycle lifetime.

CELL CHEMISTRY

NMC622

Highest energy density, lowest mfg GWP

Use-phase 50.42 g

28.07 gCO_2eq/kWh

CELL CHEMISTRY

NMC811

High energy density, low mfg GWP

Use-phase 52.36 g

29.95 gCO_2eq/kWh

CELL CHEMISTRY

LFP

Lower energy density, abundant materials

Use-phase 53.08 g

31.71 gCO_2eq/kWh

CELL CHEMISTRY

SIB

Sodium-ion, lowest use-phase impact overall

Use-phase 68.85 g

42.19 gCO_2eq/kWh

Manufacturing-phase GWP, gCO_2eq/kWh delivered (scale relative to highest)

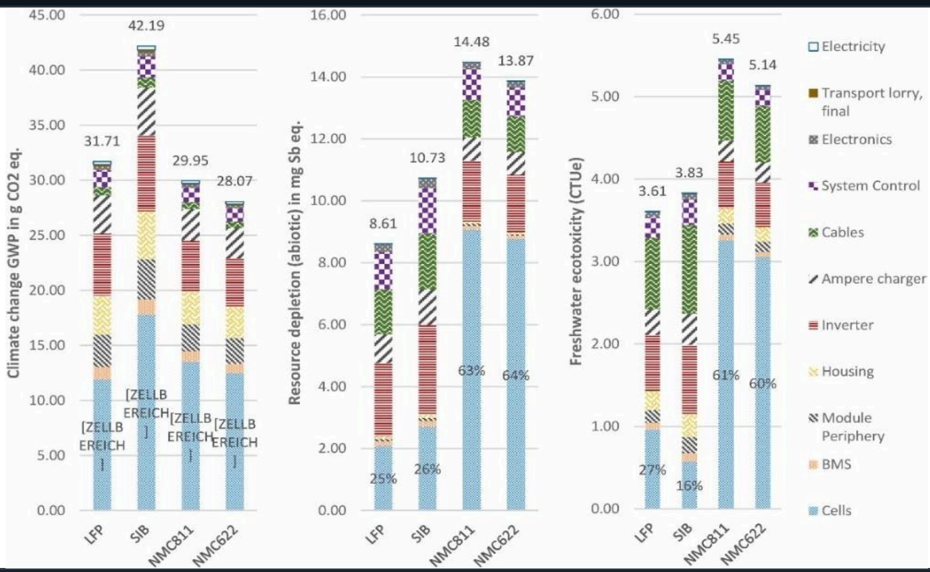

FIG.08 — Manufacturing-phase GWP, resource depletion and freshwater ecotoxicity by cell chemistry

The battery cells aren’t the whole system

Peripheral components — the inverter, system control, cabling, housing — account for **35.5% to 61.2%** of total manufacturing impact, often outweighing the cells themselves. Most published battery LCAs focus narrowly on the cells and miss this.

Recycling pays off most for NMC

up to 90% offset

End-of-life recycling can offset up to **90% of the resource-depletion impact** of NMC cell production, largely by recovering scarce cobalt and nickel. LFP cells, which use more abundant materials to begin with, see smaller absolute recycling gains — there's simply less scarce material to recover.

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05 ROADMAP

Three phases to a carbon-neutral system

Combining the findings from all three subsystems, the team mapped a phased roadmap — from near-term optimization of today's commercial technology, through scaled deployment, to a fully sustainable end state.

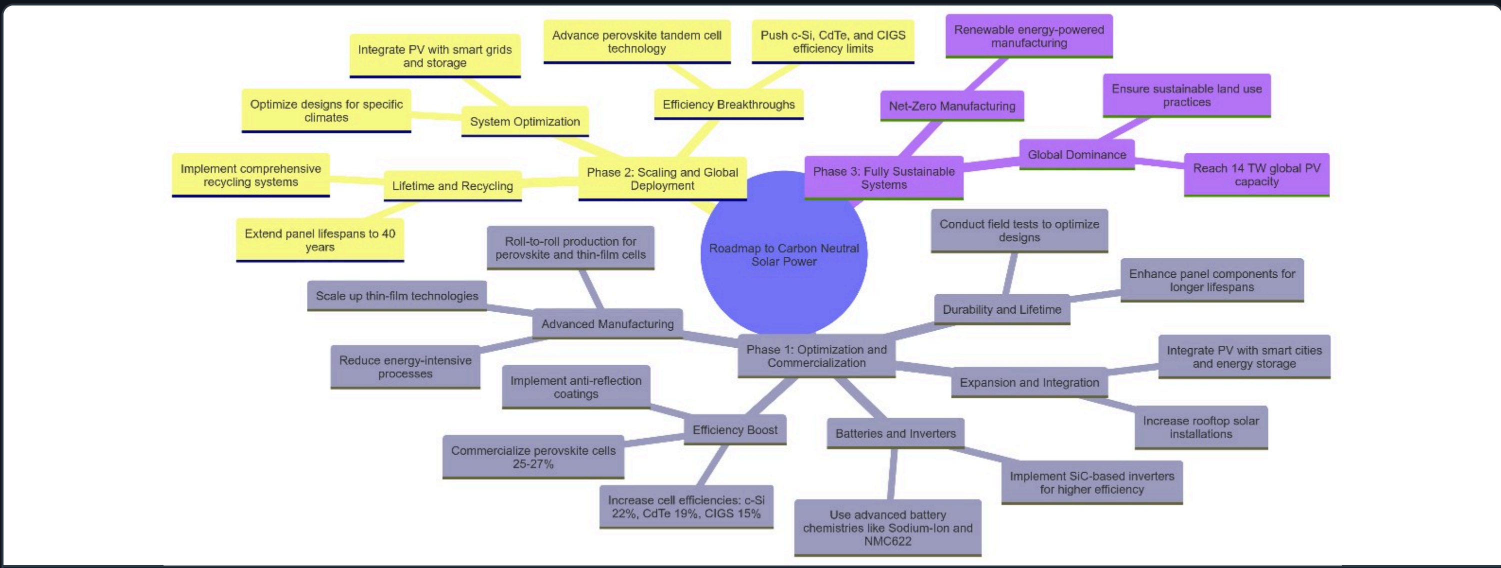

FIG.09 — Roadmap to carbon-neutral solar power, as presented to the class

PHASE 1

Optimization & Commercialization

- Boost efficiency: anti-reflection coatings, commercialize perovskite cells toward 25–27%
- Scale thin-film and roll-to-roll manufacturing to cut energy-intensive processes
- Extend panel lifespans toward 40 years; field-test durability by climate
- Adopt SiC-based inverters and sodium-ion / NMC622 battery chemistries

PHASE 2

Scaling & Global Deployment

- Push c-Si, CdTe and CIGS toward their practical efficiency limits
- Advance perovskite tandem cell technology toward field-readiness
- Implement comprehensive recycling systems and design for disassembly
- Optimize system and site-specific designs, integrate with smart grids

PHASE 3

Fully Sustainable Systems

- Shift manufacturing itself to renewable-powered, net-zero processes
- Ensure sustainable land-use practices as deployment scales
- Reach global PV capacity targets (~14 TW by 2050) on a clean grid
- Global dominance of low-carbon PV as the default energy source

06 SUMMARY

What this project demonstrates

- Life cycle assessment methodology: system boundaries, functional units, impact categories
- Component-level carbon accounting for power electronics from a bill of materials
- Cross-functional teamwork across panel, inverter and battery subsystems
- Synthesizing and reconciling variability across dozens of published LCA studies
- Translating quantitative LCA findings into a strategic, phased roadmap
- Communicating technical sustainability findings to a non-specialist audience

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